

Needles in an Image Stack: Dual Use Star Tracker Resident Space Object Streaks (DUSTstreaks)

Acquisition

RSONAR 2022
stratospheric
balloon flight
**Dual-Use Star
Tracker
(DUST)**

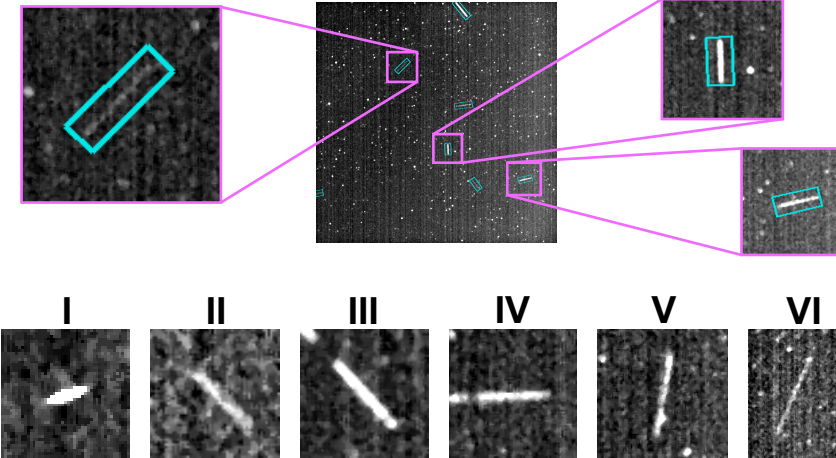


Preprocessing

Stack 27
exposures
Bias & flat
correction
Denoise &
stretch



DUSTstreak Dataset



Morphological groups

Detection Benchmark

Best **F1** scores by detector

Faster R-CNN	80.2%
RetinaNet	78.8%
FCOS	76.8%
YOLOX-tiny	76.5%

Public Release



625 images



1574 annotations



203 unique transits



model weights